

- Q.1₄₆ 100 cards are numbered from 1 to 100. The probability that the randomly chosen card has a digit 5 is:
 (A) 0.01 (B) 0.09 (C*) 0.19 (D) 0.18

[Hint: from 50 to 60, 10 fives and 9 other fives $\Rightarrow$ total 19 $\Rightarrow$ (C)]

- Q.2₄ A quadratic equation is chosen from the set of all the quadratic equations which are unchanged by squaring their roots. The chance that the chosen equation has equal roots is :
 (A*) $1/2$ (B) $1/3$ (C) $1/4$ (D) $2/3$

- Q.3₁₀ If the letters of the word "MISSISSIPPI" are written down at random in a row, the probability that no two S's occur together is :
 (A) $1/3$ (B*) $7/33$ (C) $6/13$ (D) $5/7$

- Q.4₂₉ A sample space consists of 3 sample points with associated probabilities given as $2p$, p^2 , $4p - 1$ then
 (A*) $p = \sqrt{11} - 3$ (B) $\sqrt{10} - 3$ (C) $\frac{1}{4} < p < \frac{1}{2}$ (D) none

[Hint: $p^2 + 2p + 4p - 1 = 1$ (exhaustive)
 $p^2 + 6p - 2 = 0 \Rightarrow$ (A)]

- Q.5₃₉ A committee of 5 is to be chosen from a group of 9 people. The probability that a certain married couple will either serve together or not at all is :
 (A) $1/2$ (B) $5/9$ (C*) $4/9$ (D) $2/3$

[Hint: $\frac{{}^7C_3 + {}^7C_5}{{}^9C_5}$]

- Q.6₅₃ There are only two women among 20 persons taking part in a pleasure trip. The 20 persons are divided into two groups, each group consisting of 10 persons. Then the probability that the two women will be in the same group is :
 (A*) $9/19$ (B) $9/38$ (C) $9/35$ (D) none

[Hint: $n(S)$ = number of ways in which 20 people can be divided into two equal groups

$$= \frac{20!}{10! 10! 2!}$$

$n(A)$ = 18 people can be divided into groups of 10 and 8

$$= \frac{18!}{10! 8!}$$

$$P(E) = \frac{18!}{10! 8!} \cdot \frac{10! 10! 2}{20!} = \frac{10 \cdot 9 \cdot 2}{20 \cdot 19} = \frac{9}{19} \text{ Ans]}$$

- Q.7 A bag contain 5 white, 7 black, and 4 red balls, find the chance that three balls drawn at random are all white. [Ans. $1/56$]

- Q.8 If four coins are tossed, find the chance that there should be two heads and two tails.

[Ans. 3/8]

Q.9 Thirteen persons take their places at a round table, show that it is five to one against two particular persons sitting together.

Q.10 In shuffling a pack of cards, four are accidentally dropped, find the chance that the missing cards should be one from each suit.

$$[\text{Ans. } \frac{(13)^2}{52C_4} = \frac{2197}{20825}]$$

Q.11 A has 3 shares in a lottery containing 3 prizes and 9 blanks, B has 2 shares in a lottery containing 2 prizes and 6 blanks. Compare their chances of success.

[Ans. 952 to 715]

Q.12 There are three works, one consisting of 3 volumes, one of 4 and the other of one volume. They are placed on a shelf at random, prove that the chance that volumes of the same works are all together is $\frac{3}{140}$.

Q.13 The letter forming the word *Clifton* are placed at random in a row. What is the chance that the two vowels come together?

[Ans. 2/7]

Q.14 Three bolts and three nuts are put in a box. If two parts are chosen at random, find the probability that one is a bolt and one is a nut.

[Ans. 3/5]

Q.15 There are ' m ' rupees and ' n ' ten nP's, placed at random in a line. Find the chance of the extreme coins being both ten nP's.

$$[\text{Ans. } \frac{n(n-1)}{(m+n)(m+n-1)}]$$

Q.16 A fair die is tossed. If the number is odd, find the probability that it is prime.

[Ans. 2/3]

Q.17 Three fair coins are tossed. If both heads and tails appear, determine the probability that exactly one head appears.

[Ans. 1/2]

Q.18 3 boys and 3 girls sit in a row. Find the probability that (i) the 3 girls sit together. (ii) the boys are girls sit in alternative seats.

[Ans. 1/5, 1/10]

Q.19 A coin is biased so that heads is three times as likely to appear as tails. Find $P(H)$ and $P(T)$.

[Ans. 3/4, 1/4]

Q.20 In a hand at "whist" what is the chance that the 4 kings are held by a specified player?

$$[\text{Ans. } \frac{{}^4C_4 \cdot {}^{48}C_9}{{}^{52}C_{13}}]$$